Complete Node.js Backend Development

Comprehensive Industry Curriculum — 40 In-Depth Lectures Across 6 Core Modules

TOTAL LECTURES	CORE MODULES	PRIMARY STACK	PREREQUISITES
40 Topics	6 Modules	Node / Express / Mongo	JavaScript (ES6+)

MODULE 1 — Backend & Node.js Fundamentals

Lectures 01 – 08

LEC	TOPIC	DETAILED BREAKDOWN & SUBTOPICS
01	**Backend Intro**	Frontend vs Backend, Client-Server architecture, APIs, HTTP basics, complete Backend Roadmap.
02	**Node.js Setup**	Node.js internals, V8 Engine, JS Runtime vs Browser, Installation, VS Code setup, First Node program.
03	**Node Architecture**	Deep dive into V8 engine, libuv library, Event Loop fundamentals, Non-blocking I/O paradigm.
04	**Node.js Modules**	Module systems: CommonJS (`require`/`module.exports`) vs ES Modules (`import`/`export`), interoperability.
05	**npm & Project Setup**	Managing npm, package.json structure, production dependencies vs devDependencies, custom scripts.
06	**Core Modules**	Hands-on with native modules: `fs` (file system), `path`, `os`, `url`, `crypto` for utility, and `process`.
07	**Async JavaScript**	Handling async operations: Callbacks, Promises, `async/await` patterns, and robust error handling.
08	**Event Loop & Events**	Detailed Event Loop phases, custom event emitters with `EventEmitter`, `setTimeout` / `setInterval` timing.

MODULE 2 — HTTP & Native Node Backend

Lectures 09 – 12

LEC	TOPIC	DETAILED BREAKDOWN & SUBTOPICS
09	**HTTP Fundamentals**	Understanding HTTP protocol, Request/Response cycles, Headers, Status Codes (2xx, 3xx, 4xx, 5xx).
10	**Node HTTP Server**	Building raw HTTP servers using `http.createServer()`, inspect incoming requests and formulate responses.
11	**Native Routing**	Handling routes manually, HTTP methods (GET, POST), URL parsing, route params, query strings.
12	**API Without Express**	Building a complete Mini REST API using pure Node.js to master core request lifecycle without abstractions.

Pedagogical Milestone: Module 2 ensures students thoroughly understand core Node.js server mechanics before introducing framework shortcuts like Express.js.

MODULE 3 — Express.js Complete

Lectures 13 – 20

LEC	TOPIC	DETAILED BREAKDOWN & SUBTOPICS
13	**Express.js Intro**	Express introduction, installation, setting up express server, basic routing, framework benefits.
14	**Express Routing**	Defining API routes using HTTP methods: GET, POST, PUT, PATCH, DELETE; route chaining.
15	**Request Data**	Parsing request inputs: URL Params (`req.params`), Query (`req.query`), Body (`req.body`), Headers.
16	**Middleware System**	Understanding middleware stack, Built-in, Custom application-level middleware, Third-party packages.
17	**Error Handling**	Global error-handling middleware, creating custom operational Error classes, handling uncaught exceptions.
18	**Project Architecture**	Structuring enterprise apps: Model-View-Controller (MVC) pattern, Routes, Controllers, Services, Utils.
19	**REST API Design**	Core RESTful guidelines, resource naming conventions, standardize API response envelopes (data, error, meta).
20	**API Input Validation**	Validating client inputs using Joi or Zod libraries, schema definition, custom validation error responses.

MODULE 4 — MongoDB & Mongoose Database

Lectures 21 – 28

LEC	TOPIC	DETAILED BREAKDOWN & SUBTOPICS
21	**Database Core**	SQL vs NoSQL paradigms, MongoDB architecture, Collections, BSON Documents, when to choose NoSQL.
22	**MongoDB Setup**	Cloud database setup using MongoDB Atlas, local management via MongoDB Compass, driver connections.
23	**Mongoose ORM**	Mongoose Object Data Modeling (ODM), Schema design, Model instantiation, DB connection lifecycle.
24	**MongoDB CRUD**	Complete CRUD Operations: `create()`, `find()`, `findById()`, `updateOne()`, `deleteOne()`, etc.
25	**Advanced Queries**	MongoDB Query Operators (`$gt`, `$in`, `$or`), field filtering, sorting (`sort`), pagination (`skip`/`limit`).
26	**Relationships**	Data modeling: Normalized (References via ObjectIds) vs Embedded Documents, relational joins using `.populate()`.
27	**Indexing & Perf**	Database Indexing strategy, Single & Compound indexes, Unique constraints, performance optimization.
28	**Aggregation Pipeline**	Complex analytics queries: Aggregation stages (`$match`, `$group`, `$lookup`, `$project`, `$sort`, `$unwind`).

MODULE 5 — Authentication, Authorization & Security

Lectures 29 – 35

LEC	TOPIC	DETAILED BREAKDOWN & SUBTOPICS
29	**Auth Fundamentals**	Authentication vs Authorization, Session-Cookie based auth vs Stateless JSON Web Tokens (JWT).
30	**User Registration**	User registration flow, login endpoint, email/password validation schemas, duplicate user handling.
31	**Password Hashing**	Password security standards, salt rounds, cryptographic hashing using `bcrypt`, password verification.
32	**JWT Deep Dive**	JWT structure (Header, Payload, Signature), issuing Access Tokens, token verification and expiration.
33	**Protected Routes**	Authentication middleware, extracting Bearer tokens, attaching authenticated user payload to `req.user`.
34	**RBAC System**	Role-Based Access Control: defining roles (Admin, Manager, Employee), permission-checking middleware.
35	**Refresh & Logout**	Advanced Auth: Short-lived access tokens, Refresh Token rotation in HTTP-Only cookies, secure Logout.

MODULE 6 — Production Features, Testing & Deployment

Lectures 36 – 40

LEC	TOPIC	DETAILED BREAKDOWN & SUBTOPICS
36	**File Uploads**	Handling multipart data with Multer, image filtering/validation, cloud storage integration via Cloudinary.
37	**Email & OTP Service**	Sending transactional emails via Nodemailer, OTP generation/verification for Password Reset workflows.
38	**Production Security**	API Security best practices: CORS config, Helmet HTTP headers, Rate Limiting, Data Sanitization, `.env`.
39	**Testing & Docs**	API testing workflows with Postman, automated Integration testing basics, OpenAPI/Swagger documentation.
40	**Deployment & PM2**	Git version control, Cloud Deployment (Render/AWS/VPS), production MongoDB Atlas setup, PM2 process management & logs.

🎓 Capstone Course Project

Throughout Module 5 & 6, students build a **Production-Ready Multi-Role SaaS REST API** featuring JWT Auth with Refresh Tokens, Role-Based Access Control, File Uploads, Email Notification workflows, and live deployment with process management.